

WASP-52b

R-1636

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_210921 · created 2026-08-02T06:26:32+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459478.7106 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.207 +/- 0.061
Transit depth (Rp/Rs)^2	4.29 +/- 2.51 [%]
Orbital Inclination (inc)	84.11 +/- 2.75
Airmass coefficient 1 (a1)	1.17 +/- 0.7
Airmass coefficient 2 (a2)	-0.18 +/- 0.77
Scatter in the residuals of the lightcurve fit is	56.36 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	5.75
Transit Duration (day)	0.063 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.207 ± 0.061 vs 0.1646 (deviation 0.68σ)
Duration fit vs published	0.063 d vs 0.0758 d (deviation 0.46σ)
Mid-time O–C	-14.6 min
Depth SNR	3.3
Residual scatter	56.36 %

Light curve

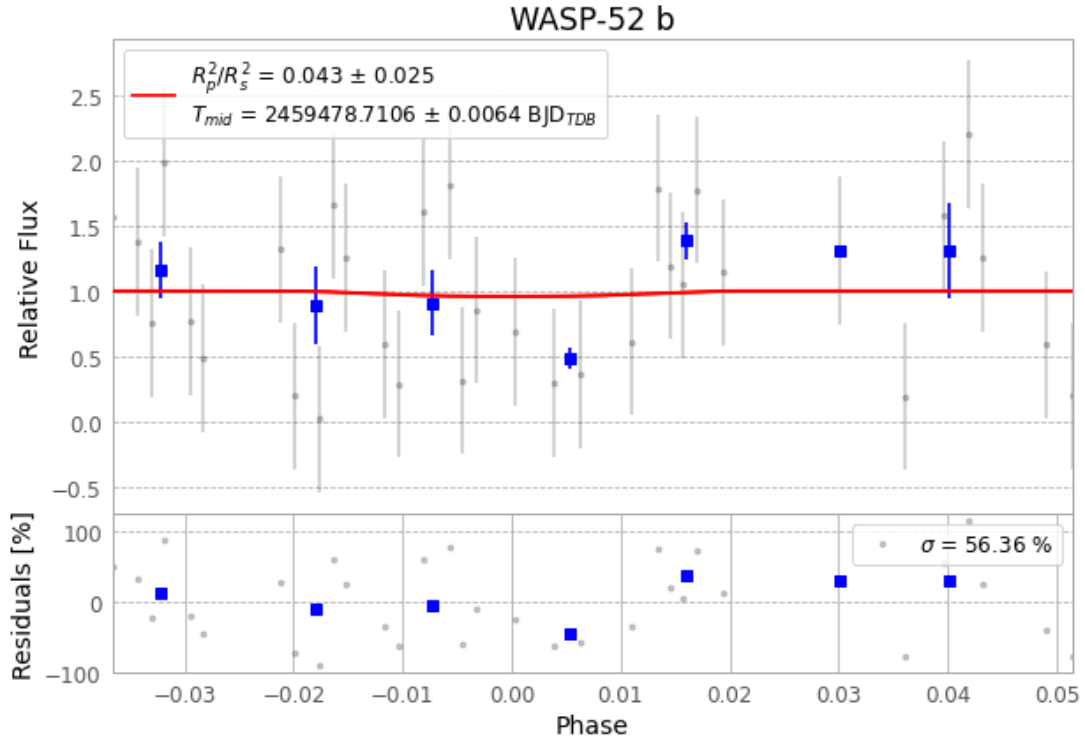


Figure 1 — FinalLightCurve_WASP-52 b_2021-09-20.png (SHA-256 3d801068f6197504...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 71], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 88ab1858f99d1ee5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52210921032113.FITS → WASP-52210921070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-210921.log (a4c4ea419f39...), FinalLightCurve_WASP-52 b_2021-09-20.png (3d801068f619...), FinalLightCurve_WASP-52 b_2021-09-20.csv (38a56d9475a8...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 06:26Z.